🚀 TECHLIFT – STAGE 3

🧩 TECH MISSION: CRACK • DEBUG • PREDICT • PRESENT

Duration: 60 minutes
Participants: Qualified teams from Stage 2
Team size: 2–3
Mode: Team Activity
Total: 100 marks

This will be much more fun than another coding round.

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1. How the Activity Works

Each team receives a **TECH MISSION PACK** containing four challenges.

🔵 **Mission 1 — Predict the Output**

Students look at short Python programs and predict what they will print.

🟠 **Mission 2 — Debug It**

Students receive programs containing errors and must identify and correct them.

🟢 **Mission 3 — Tech Puzzle**

Students solve logical/technical puzzles.

🔴 **Mission 4 — Tech Pitch**

Students receive a simple real-world technology problem and prepare a **2-minute solution presentation**.

So the progression is:

Think → Find → Solve → Explain

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2. 60-MINUTE TIMELINE

Time	Activity
0–5 min	Rules + distribution of Tech Mission Pack

Time	Activity
5–15 min	Mission 1 – Output Prediction
15–30 min	Mission 2 – Debugging
30–45 min	Mission 3 – Tech Puzzle
45–55 min	Mission 4 – Prepare Solution
55–60 min	2-minute presentations

Important

Don't give each team 2 minutes separately if you have too many teams.

For Stage 3, I recommend qualifying around **5–8 teams** from Stage 2.

That makes the presentation section manageable.

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3. MARK DISTRIBUTION

Let's map the activity directly to your official Stage 3 criteria.

Criterion	Marks
Concept Understanding	25
Accuracy	25
Problem-Solving	20
Creativity / Participation	15
Presentation / Communication	15
TOTAL	**100**

🔵 MISSION 1 — OUTPUT PREDICTION

Duration

10 minutes

Marks

20

Give teams **4 small Python snippets**.

They don't have to write code.

They simply predict the output.

Question 1

```
x = 10
y = 5

print(x + y)
print(x * y)
```

Expected Answer

15

50

Marks: 5

Question 2

```
a = 10

if a > 5:
    print("YES")
else:
    print("NO")
```

Expected Answer

YES

Marks: 5

Question 3

```
numbers = [2, 4, 6, 8]


print(numbers[0])
print(numbers[-1])
```

Expected Answer

2

8

Marks: 5

Question 4

```
for i in range(1, 5):
    print(i)
```

Expected Answer

1

2

3

4

Marks: 5

🔴 MISSION 2 — DEBUGGING

Duration

15 minutes

Marks

30

This is the main activity.

Give each team **3 buggy programs**.

They must:

1. Identify the error
2. Explain the error
3. Correct the code

Debugging Challenge 1

Given Code

```
name = input()

if name == "python"
   print("Correct")
else:
   print("Wrong")
```

Task

Find and correct the error.

Error

Missing : after the if condition.

Correct Code

```
name = input()

if name == "python":
   print("Correct")
else:
   print("Wrong")
```

Marks

10

- Identify error – 3
- Explain – 2

- Correct code – 5

Debugging Challenge 2

Given Code

```
numbers = [10, 20, 30, 40]

total = 0

for n in numbers:
    total = n

print(total)
```

Expected Output

100

What's wrong?

The program replaces total every time instead of adding to it.

Incorrect

```
total = n
```

Correct

```
total += n
```

Correct Code

```
numbers = [10, 20, 30, 40]

total = 0

for n in numbers:
    total += n
```

```
print(total)
```

Marks

10

Debugging Challenge 3

Given Code

```
age = input()

if age >= 18:
    print("Eligible")
else:
    print("Not Eligible")
```

Problem

input() returns a string.

The program is trying to compare a string with an integer.

Correct Code

```
age = int(input())

if age >= 18:
    print("Eligible")
else:
    print("Not Eligible")
```

Marks

10

🟢 MISSION 3 — TECH PUZZLE

Duration

15 minutes

Marks

25

This is where we make the activity more exciting.

Give each team a **Puzzle Card**.

They can solve the puzzles collaboratively.

Puzzle 1 — The Missing Number

Find the next number:

2, 6, 12, 20, 30, ?

Answer

42

Explanation

The pattern is:

$1 \times 2 = 2$

$2 \times 3 = 6$

$3 \times 4 = 12$

$4 \times 5 = 20$

$5 \times 6 = 30$

$6 \times 7 = 42$

Marks: 5

Puzzle 2 — Binary Challenge

Convert:

1010

from binary to decimal.

Answer

10

Because:

1×8 + 0×4 + 1×2 + 0×1

= 8 + 2

= 10

Marks: 5

Puzzle 3 — Logic Challenge

Three switches are outside a room.

Inside the room is one bulb.

Only one switch controls the bulb.

You can operate the switches however you want, but you may enter the room **only once**.

How do you determine which switch controls the bulb?

Expected Solution

1. Turn ON switch 1 for a few minutes.
2. Turn switch 1 OFF.
3. Turn switch 2 ON.
4. Enter the room.

Then:

- Bulb ON → Switch 2
- Bulb OFF but warm → Switch 1
- Bulb OFF and cold → Switch 3

Marks: 10

Puzzle 4 — Technology Riddle

I can store your files.
I have no keyboard.
I can fit in your pocket.
I use memory to keep your data.
What am I?

Answer

USB Flash Drive / Pen Drive

Marks: 5

🔴 MISSION 4 — TECH PITCH

This is the most important part for making Stage 3 feel like a real **activity** rather than just another test.

Preparation Time

10 minutes

Presentation

2 minutes per team

Marks

25

Give Each Team ONE Problem Card

Don't give every team the same problem.

Prepare different cards and randomly distribute them.

💡 Problem Card 1 — Smart Attendance

Your college wants to reduce the time spent taking attendance.

Design a simple technology-based solution that can make attendance faster and easier.

Team must explain:

- Problem

- Proposed solution
- How technology is used
- Benefits

💡 Problem Card 2 — Smart Canteen

Students spend too much time waiting in the college canteen.

Design a technology-based solution to reduce waiting time.

💡 Problem Card 3 — Lost & Found

Students frequently lose ID cards, books, USB drives and other items.

Design a simple digital Lost & Found system.

💡 Problem Card 4 — Lab Equipment

Students sometimes don't know whether a particular computer/lab equipment is working or already occupied.

Design a simple technology solution.

💡 Problem Card 5 — Campus Navigation

New students often struggle to find classrooms, laboratories and offices.

Design a technology-based campus navigation solution.

💡 Problem Card 6 — Study Assistant

First-year students often struggle to organize notes, assignments and study schedules.

Design a simple technology solution to help them.

🎤 PRESENTATION FORMAT

Give each team exactly **2 minutes**.

Tell them to follow:

30 seconds

Problem

"What is the problem?"

45 seconds

Solution

"What are you proposing?"

30 seconds

How it works

"How will technology solve it?"

15 seconds

Impact

"What benefit does it provide?"

🏆 **PRESENTATION MARKS**

Area	Marks
Understanding the problem	5
Proposed solution	5
Creativity	5
Technical thinking	5
Communication	5
TOTAL	**25**

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4. FINAL SCORE

Here's how we can convert the activities into the official 100-mark score.

Activity	Marks
Output Prediction	20
Debugging	30
Tech Puzzle	25
Tech Pitch	25
TOTAL	**100**

Then the judging team evaluates the overall performance using the official criteria:

Concept Understanding – 25%
Accuracy – 25%
Problem-Solving – 20%
Creativity/Participation – 15%
Presentation/Communication – 15%.

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5. MAKE IT MORE FUN 🔥

Instead of simply giving them a question paper, prepare a **TECH MISSION ENVELOPE** for every team.

For example:

TEAM TL-01

📦 **TECH MISSION PACK**

Inside:

Card 1: Output Prediction
Card 2: Debugging
Card 3: Tech Puzzle
Card 4: Tech Pitch

At the beginning announce:

"You have 60 minutes to complete the TECH MISSION. Your team must THINK, DEBUG, CRACK and PRESENT your solution."

This makes the round feel like an actual competition rather than an examination.

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6. MATERIALS YOU NEED

For Stage 3, prepare:

For every team

- 1 Tech Mission envelope
- 1 answer sheet
- 1 problem card
- 2–3 rough sheets
- Pen/pencil
- Team ID card

Organizers

- Timer
- Stopwatch
- Puzzle cards
- Debugging sheets
- Output prediction sheets
- Presentation evaluation sheets
- Score sheets
- Projector
- Laptop
- Bell/announcement system

The general event documentation also identifies activity cards, timer/stopwatch, answer/evaluation sheets, projector, computer setup and backup materials as event requirements.

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7. QUALIFICATION FROM STAGE 2 → STAGE 3

Because TECHLIFT may have around **80–100 students**, you shouldn't bring everyone into this activity round.

I recommend:

Stage 1 → Stage 2

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Stage 2 → Top 6–8 teams

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Stage 3 → Final Winner

For example:

Team	Stage 2 Result	Stage 3
TL-01	Qualified	✅
TL-04	Qualified	✅
TL-07	Qualified	✅
TL-09	Qualified	✅
TL-12	Qualified	✅
TL-15	Qualified	✅
TL-18	Qualified	✅
TL-21	Qualified	✅

This also makes the **presentation component realistic within one hour**.

🔥 Final TECHLIFT Structure

You now have a very clear progression:

STAGE 1 🧠

Technical Quiz

20 questions
15 minutes

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STAGE 2 💻

Beginner Coding Challenge

5 problems
60 minutes

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STAGE 3 🧩

TECH MISSION

Output Prediction → Debugging → Puzzle → Tech Pitch

60 minutes

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🏆 **TECHLIFT WINNER**

This fits the intended progression in your event documentation: **Technical Quiz → Beginner Coding Challenge → Tech Activity**, moving from fundamentals to programming/problem-solving and finally practical activity and communication.

The next thing I'd prepare is the actual printable Stage 3 material: the **Tech Mission Question Sheet, 6 Problem Cards, Answer Key, Judge Evaluation Sheet, and Final Score Sheet** so you can print them and use them directly on 27 October.